

GP-AI-Agent

A local-first AI agent that spends tokens only when a stronger model is needed.

THE IDEA

Route locally.
+Escalate safely.

Task arrives



Learned router

LOCAL

REMOTE

Local Qwen

Fireworks

Strong models are valuable - but using them for every task is wasteful.

01

Cost grows with every prompt

A simple question can trigger the same external-model path as a difficult coding task.

02

Latency is unnecessary for safe tasks

Some tasks can be solved locally and verified deterministically in the container.

TRACK 1 CONSTRAINTS

4 GB RAM

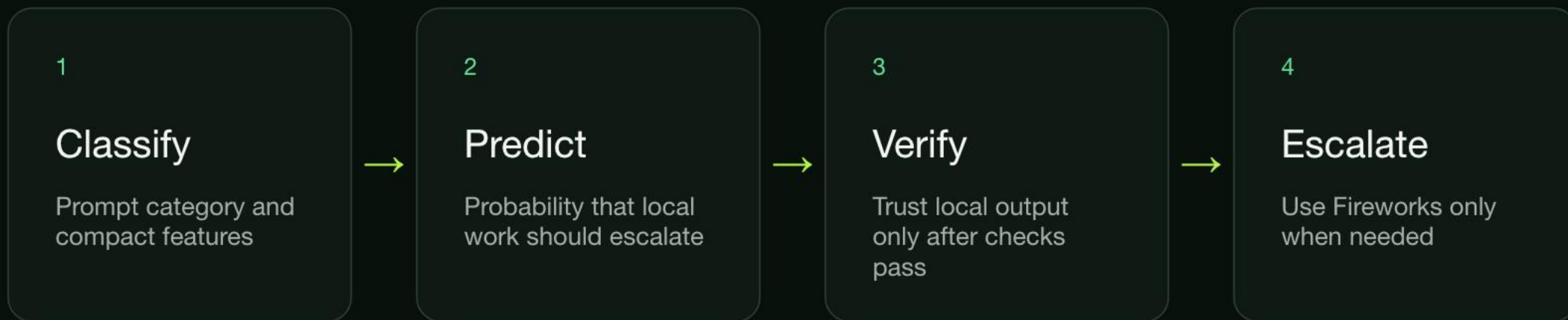
2 vCPU

< 10 min total runtime

The agent must use a small local model, a disciplined budget, and a reliable fallback.

GP-AI-Agent learns when the local path is trustworthy.

The router makes a free decision in the browser and container before any hosted API call.



A compact router, a local solver, and a safety gate work as one system.

01 LEARNED ROUTER

Hashed unigram +
bigram
logistic classifier

Trained from 360 measured
local outcomes.
Calibrated threshold plus
category policy.

02 LOCAL SOLVER

Qwen 2.5 1.5B
Q4_K_M via
llama.cpp

CPU-only inference sized for
the 4 GB / 2 vCPU environment.

03 SAFETY + FALLBACK

Verify locally.
Escalate when
needed.

Arithmetic, code, logic, and
format checks protect the final
answer.

Local answers use zero Fireworks routing tokens.

The optimized policy cuts expected API tokens while protecting reliability.

9,685

previous API tokens

1,824

v11 expected API tokens

97.14%

v11 projected accuracy

19 / 19

rehearsal answers completed

Validation evidence

80 tasks answered in 207.4 seconds • 45 remote calls succeeded • 0 fallbacks • 130 / 130 tests passing

Projected accuracy and token estimates come from the held-out router profile; rehearsal figures are measured end-to-end.

The result is a practical agent that is fast, budget-aware, and safe by design.

LOCAL-FIRST DOES NOT MEAN LOCAL-ONLY

It means every task gets the cheapest reliable path - with a stronger model ready when the task deserves it.

LIVE DEMO

[gp-ai-agent-1
.vercel.app](https://gp-ai-agent-1.vercel.app)

Route locally. Spend only when needed.